



The Effect of Aerofoil Design on Aircraft Performance

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Introduction

Context

An autonomous micro air vehicle (MAV) will be designed for a future smart city. The MAV is to be powered by an electric motor and therefore both the range and duration are limited by the available battery life. It is supposed to fly to the scene of, for example, a fire, and carry out a detailed investigation of the location with video images.

Objective

This report shall outline the selection process for a suitable aerofoil for this MAV wing design. XFOIL shall be used to predict the aerodynamic performance of aerofoils alongside analytical solutions for parameters such as aspect ratio.

The key design goals for this selection are minimising drag, achieving efficient lift, and maintaining flight at low Reynolds numbers.

Design Specification

- Altitude is assumed to be sea level
- The MAV shall be powered by a small motor and battery
- MAV speed (v) is approximated at 11 m/s
- MAV mass (m) is 0.51 kg, resulting in a required lift (L) of 5 N
- The MAV wing area (S) shall be 0.14 m²
- The 3D wing lift coefficient is assumed to be 90% of the 2D aerofoil performance
- It is assumed that the thickness distribution follows the NACA 4-digit profile.
- It is assumed the MAV has a conventional rectangular wing design
- Oswald efficiency factor of the wing shall be taken as 0.7 [1]
- It is assumed that only the wing design will contribute to the aerodynamic forces and moments from the body.

Approach

The AR will be chosen by evaluating its influence on lift-induced drag and the effects of low Reynolds number flows, typical for MAVs operating at low speeds. An appropriate AR will be selected to minimise drag and ensure efficient flight by analysing the relationship between AR and these performance factors.

A range of NACA 4-digit aerofoil profiles with varying thickness and camber will be analysed using XFOIL software. The goal is to identify the aerofoil with the lowest drag coefficient at the MAV's design conditions while providing sufficient lift. The 3D wing performance can then be calculated assuming a 10% lift coefficient drop due to 3D effects.

Finally, the required thrust for the MAV will be calculated based on the drag characteristics of the chosen aerofoil. The thrust estimation will include the span efficiency factor to account for the wing's aspect ratio and aerodynamic performance, ensuring the MAV can maintain stable flight at its design speed of 11 m/s.

Aspect Ratio Selection

Given the MAV flies at low speeds, the main factors to consider when selecting an aspect ratio are the induced drag, structural feasibility, and low Reynolds number effects at low chord lengths. Starting with the induced drag, we can use the following equations:

$$C_{Di} = \frac{C_L^2}{\pi AR e} \quad (\text{Eq. 1})$$

$$C_L = \frac{2L}{\rho V^2 S} \quad (\text{Eq. 2})$$

Eq. 2 can be used to find the coefficient of lift required for steady-level flight **$C_L = 0.48$**

Using this value alongside the assumption the Oswald efficiency factor is 0.7, a relationship can be found between the induced drag and the aspect ratio

$$C_{Di} = \frac{0.48^2}{\pi AR e} = \frac{0.48^2}{\pi AR \times 0.7} = \frac{0.105}{AR} \quad (\text{Eq. 3})$$

Using this formula it can be shown that the higher the aspect ratio, the lower the drag. However, there are limitations to high aspect ratios, such as structural limits, MAV storage, and the wing's flow characteristics approaching lower Reynolds numbers. The next best course of action is to look at existing aircraft for reference and select a case study to justify a selection.



Figure 1.0 - VolantexRC Phoenix V2 2000mm Glider [2]



Figure 1.1 - MicroMax Pocket F3F RTF Glider [2]

Figure 1 shows two RC Gliders capable of carrying 500g, which matches the capabilities of the desired MAV. These have an average aspect ratio of **$AR = 9.5$** , which is selected for this MAV.

Eq. 4 is the equation for the Reynolds number with the characteristic dimension (wing chord) written in terms of the surface area and the aspect ratio.

$$Re = \frac{\rho V \left(\frac{\sqrt{S}}{\sqrt{AR}} \right)}{\mu} \quad (\text{Eq. 4})$$

Substituting design specifications into Eq. 4 returns a Reynolds number of **$Re = 90,375$** . This means the glider will be flying in a laminar regime that is good for stability and low drag.

Aerofoil Selection

As calculated in Eq. 2, the optimal c_L for steady-level flight is $C_L = 0.48$ for 3D wings, but this value needs to be increased by 10% for 2D aerofoils. This results in a new value of the 2D $c_L = 0.53$.

To narrow down the range of aerofoils to test, certain ranges of aerofoils are neglected in this study:

- X400 to X405 neglected for structural infeasibility and XFOIL convergence issues
- X415 to X440 neglected for excessive thickness leading to increased form drag and weight

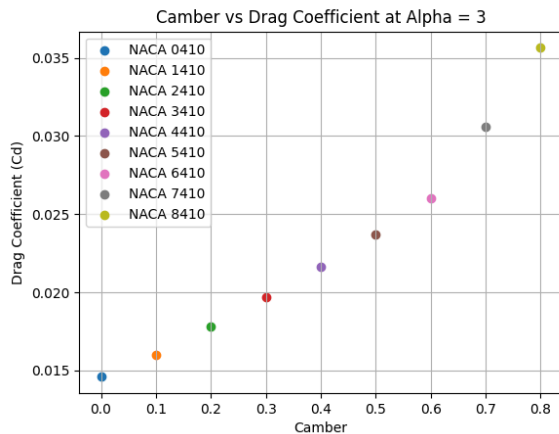


Figure 2.0 - Camber vs Drag Coefficient at AoA = 3

The camber is the first thing to be tested, this test aims to narrow down to a range of cambers. Figure 2.0 shows the drag coefficient across different cambers, with an angle of attack of 3° .

This relationship highlights that after 44XX, the drag coefficient increases beyond feasibility. Hence, the range of cambers to be tested is 04XX to 34XX.

Candidate aerofoils:

0408, 0410, 0412, 1408, 1410, 1412, 2408, 2410, 2412, 3408, 3410, 3412

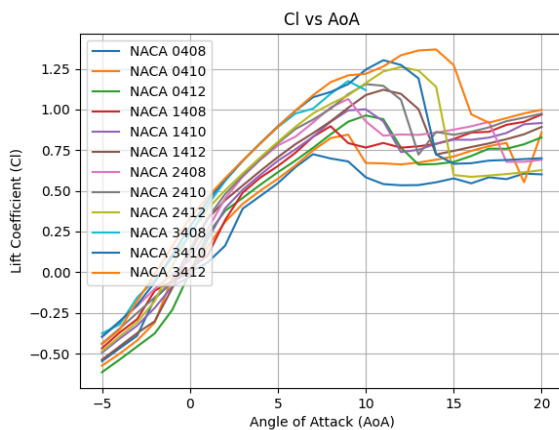


Figure 3.0 - Lift Coefficient vs Angle of Attack

Next, the lift performance is calculated to select aerofoils with desirable stall characteristics.

Figure 3 demonstrates that most aerofoils can reach the desired c_L of 0.53 at an AoA of 3° . The aerofoils that don't are rejected.

Aerofoils with a stall angle of less than 10° are rejected as they would increase the stall risk.

Candidate aerofoils:

0408, 0410, 0412, 1408, 1410, 1412, 2408, 2410, 2412, 3408, 3410, 3412

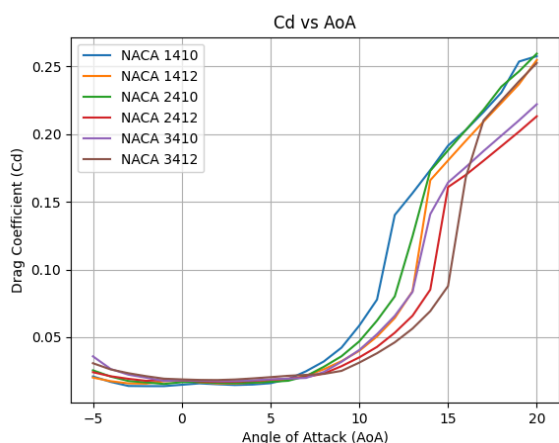


Figure 4.0 - Drag Coefficient vs Angle of Attack

The next stage of the selection process is to select the aerofoil with the least drag.

Figure 4 shows the drag reduces as thickness and camber increases for the range of aerofoils tested.

The aerofoils with the most drag are rejected to maintain a more efficient steady level flight.

Candidate aerofoils:

0408, 0410, 0412, 1408, 1410, 1412, 2408, 2410, 2412, 3408, 3410, 3412

The final selection process involves the drag polar plot (c_L vs c_D) which can be used to find the efficiency and stall characteristics of the remaining aerofoils around the operating point.

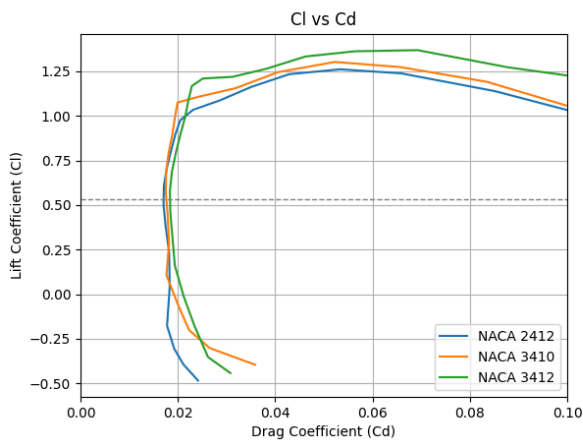


Figure 5.0 - Drag polar plot

As shown in Figure 5, the chosen aerofoils are similar around the operating point, however the stall characteristics vary. It is desirable to have a gradual stall and the lowest drag around the operating point.

For this reason, **NACA 2412** is selected.

This aerofoil is the most efficient, has good stall characteristics, and is structurally feasible for the MAV specified. Also, it is suitable for manoeuvres as the aerofoil can provide lift up to $c_L \approx 1.0$ before stalling the aircraft.

Thrust Analysis

To evaluate the thrust required for steady-level flight, the simulation results and Eq. 5 can be used.

$$D = \frac{1}{2} \rho v^2 S C_d \quad (\text{Eq. 5})$$

$$C_d = C_{d0} + C_{di} \quad (\text{Eq. 6})$$

Since thrust is equal to drag at steady level flight, this gives a reasonable estimate of the thrust required. Using Figure 5, the drag polar at the operating point can be evaluated at $C_{D0} = 0.017$. Substituting this into Eq. 6 along with $C_{Di} = 0.011$ (Eq. 3) yields $C_D = 0.028$.

Finally, using Eq.5 the thrust can be calculated to be **T = 0.29N**.

Summary of Findings

Aspect Ratio Selection

- Larger aspect ratios are more efficient, but structural limitations restrict choice
- Gliders of a similar weight class have an average AR of 9.5 which was selected for this MAV

Aerofoil Selection

- Aerofoils that were too thick or too thin were neglected from the study
- Analysis of camber vs drag demonstrated low cambers performed better
- Analysis of lift coefficient rejected most thin and flat aerofoils
- Analysis of drag coefficient rejected more thin and low-camber aerofoils
- Analysis of drag polar revealed NACA 2412 is best due to its stall characteristics and efficiency

Thrust Analysis

- The thrust required to maintain steady level flight is small enough to be provided by a small propeller and motor, which is specified in the design.

References

[1] NASA Glenn Research Center. "Induced Drag Coefficient." *Beginners Guide to Aeronautics*, NASA, 27 July 2023, Accessed 24 Oct 2024.

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[2] Hobby RC. "VolantexRC Phoenix V2 2000mm Glider (PNP)"

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[3] HyperFlight.co.uk. "MicroMax Pocket F3F RTF including servos and wing, tail & fuselage bags"

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